Architecture Implementation Support Techniques

Module 14

Module Objectives

- Obtain an understanding of the following techniques provided by TOGAF to assist in Architecture Development
  - Managing Interoperability Requirements
  - Business Transformation
  - Risk Assessment
  - Capability Planning

Interoperability

- Interoperability is "***the ability to share information and services***"
- TOGAF provides techniques for
  - **Defining** interoperability
  - **Refining** interoperability
  - Determining interoperability **requirements**
- The determination of interoperability occurs throughout the ADM cycle

Interoperability and the ADM

The **determination of interoperability** occurs throughout ADM

- **Architecture Vision**: the nature and security considerations of information and service exchanges are found using **business scenarios**
- **Business Architecture:** information and service exchanges are defined in business terms
- **Data Architecture:** the content of information exchanges is detailed using the corporate data and/or information exchange **model**
- **Application Architecture:** the way applications are to share information and **services** is specified
- **Technology Architecture:** appropriate **technical mechanisms** to permit information and service exchanges are specified
- **Opportunities & Solutions:** actual solutions are selected
- **Migration Planning:** interoperability is implemented logically

Examples

Phase B: Inter-stakeholder Information Interoperability Requirements (Using degrees of information interoperability) © The Open Group							
Stakeholders	A	B	C	D	E	F	G
A		2	3	2	3	3	3
B	2		3	2	3	2	2
C	3	3		2	2	2	3
D	2	2	2		3	3	3
E	4	4	2	3		3	3
F	4	4	2	3	3		2
G	2	2	3	3	3	3	

Phase C: Inter-system Interoperability Requirements © The Open Group							
	System A	System B	System C	System D	System E	System F	System G
System A		2A	3D	2B	3A	3A	3B
System B	2E		3F	2C	3A	2B	2C
System C	3E	3F		2B	2A	2A	3B
System D	2B	2B	2B		3A	3A	3B
System E	4A	4B	2B	3A		3B	3B
System F	4A	4A	2B	3B	3A		2D
System G	2B	2B	3A	3A	3B	3B	

Interoperability requirements and solutions

- The architect must ensure that there are no **interoperability conflicts**, especially if re-using existing SBBs or using COTS which have their own business processes and information architectures
- Changes to the business processes will be the most difficult
- The **workflow** between the various systems must also be taken into account
- The enterprise architect must also ensure that any changes to the business interoperability requirements is agreed by the business architects and sponsors in a revised **Statement of Architecture Work**

Interoperability requirements and solutions

To find interoperability constraints consider

- the **Architecture Vision**
- the **Target Architecture**
- The **Implementation Factor Assessment and Deduction** matrix
- The **Consolidated Gaps, Solutions and Dependencies** matrix

Business Transformation Readiness Assessment

- EA often involves considerable change
- Understanding the readiness of an organization to **accept change**, **identifying the issues**, and dealing with them in the Implementation and Migration Plans is key to successful architecture transformation in **Phases E and F**. An initial assessment is carried out in **Phase A**
- This is a **joint effort** between corporate staff, lines of business and IT planners

Business Transformation Readiness Assessment

Recommended activities when assessing readiness

1. Determine the **readiness factors**
2. Present the readiness factors using **maturity models**
3. Assess the readiness factors and determine the **readiness factor ratings**
4. Assess the risks for each readiness factor and identify **mitigating risks**

Work these actions into **Phase E** and **F** Implementation and Migration Plan

Readiness Factors

- **Vision**
- Desire, Willingness and Resolve
- Need, **Business Case**, Funding
- Sponsorships and Leadership
- Governance, Accountability
- **Workable Approach** and Execution Model
- **IT/Enterprise Capacity to Execute**
- Enterprise Ability to Implement and Operate

Assess the Readiness Factors

Business Transformation Readiness Assessment - Maturity Model					© The Open Group
Factor 2: Need for Enterprise Information Architecture			**Class**	Organizational Context	
			BTEP Readiness Factor	YES	
Definition	There is recognition by the organization that information is a strategic corporate asset requiring stewardship. There is also recognition that the data is not universally understandable, of requisite quality, and accessible.				
Maturity Model Levels					
0 Not defined	**1 Ad Hoc**	**2 Repeatable**	**3 Defined**	**4 Managed**	**5 Optimized**
Information is not recognized as an asset. There is no clear stewardship of data.	Data Management (DM) concepts are intuitively understood and practiced on an *ad hoc* basis. Stewardship of the data is informal. Data is recognized by certain internal experts and senior management as being of strategic importance to the organization. Focus is primarily on technically managing redundant data at the applications level.	Many parts of the organization value information/data as a strategic asset. Internal DM experts maintain clear lines of responsibility and stewardship of the data, organized along lines of business and at all senior levels. Staff put into practice DM principles and standards in their daily activities.	Data is recognized as a strategic asset in most parts of the organization, and throughout most levels from operations to senior management. Resources are committed to ensuring strong stewardship of data at the lower management and information expert levels.	Data is recognized as a strategic asset in all parts of the organization, and throughout most levels from operations to senior management. Resources are committed to ensuring strong stewardship of data at the senior management and information expert levels.	Data is treated in all levels throughout the organization as a strategic asset to be exploited and re-used. Data products and services are strongly integrated with the management practice of the organization. All staff are empowered and equipped to take stewardship of information, and are seen as "knowledge workers".
				Recommended Target State	

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Readiness Factor Rating

Business Factor Assessment Summary				© The Open Group
Ser	**Readiness Factor**	**Urgency**	**Readiness Status**	**Degree of Difficulty to Fix**
1	Vision			
2	Desire/willingness/resolve			
3	Need			
4	Business case			
5	Funding			
6	Sponsorship and leadership			
7	Governance			
8	Accountability			
9	Workable approach and execution model			
10	IT capacity to execute			
11	Departmental capacity to execute			
12	Ability to implement and operate			

Readiness Factor Risks & Actions

- Assess each factor using **Risk Management** techniques
- Identify a series of improvement actions
- **Incorporate** into the Implementation and Migration Plan

Risk Management

- A technique used to **mitigate risk** when implementing an architecture project
- It is important to **identify**, **classify**, and **mitigate** these risks before starting so that they can be tracked throughout the transformation effort

Risk Management in the ADM

There are two levels of risk that should be considered

1. **Initial Level of Risk:** Risk categorization prior to determining and implementing mitigating actions
2. **Residual Level of Risk:** Risk categorization after implementation of mitigating actions

The process for risk management is

- Risk **classification** & **identification**
- Initial risk **assessment**
- Risk **mitigation** and residual risk assessment
- Risk **monitoring**

Risk Management in the ADM

- Risks are identified in **Phase A** as part of the initial Business Transformation Readiness Assessment
- The risk identification and mitigation assessment worksheets are maintained as governance artifacts and are kept up-to-date in **Phase G** where risk monitoring is conducted
- Implementation governance can **identify critical risks** that are not being mitigated and might require another full or partial ADM cycle

Initial Risk Assessment - Effect

- The initial risk assessment is done by classifying risks with respect to **effect** and **frequency**
- Effect can be assessed as
  - **Catastrophic**: critical financial loss that could result in bankruptcy
  - **Critical**: serious financial loss in more than one line of business leading to a loss in productivity and no ROI
  - **Marginal**: minor financial loss in a line of business and a reduced ROI on the IT investment
  - **Negligible**: minimal impact on services and/or products

Initial Risk Assessment - Frequency

Frequency can be assessed as

- **Frequent:** Likely to occur very often and/or continuously
- **Likely**: Occurs several times over the course of a transformation cycle
- **Occasional**: Occurs sporadically
- **Seldom**: Remotely possible and would probably occur not more than once in the course of a transformation cycle
- **Unlikely**: Will probably not occur during the course of a transformation cycle

Initial Risk Assessment

The assessments of effect and frequency can then be combined

- **Extremely High Risk (E)**: the transformation will most likely fail with severe consequences
- **High Risk (H)**: Significant failure of parts of the transformation resulting in certain goals not being achieved
- **Moderate Risk (M)**: Noticeable failure of parts of the transformation, threatening the success of some goals
- **Low Risk (L)**: Some goals won't be wholly successful

Risk Classification Scheme

Corporate Risk Impact Assessment					
	Frequency				
Effect	Frequent	Likely	Occasional	Seldom	Unlikely
Catastrophic	E	E	H	H	M
Critical	E	H	H	M	L
Marginal	H	M	M	L	L
Negligible	M	L	L	L	L

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Risk Identification and Mitigation Worksheet

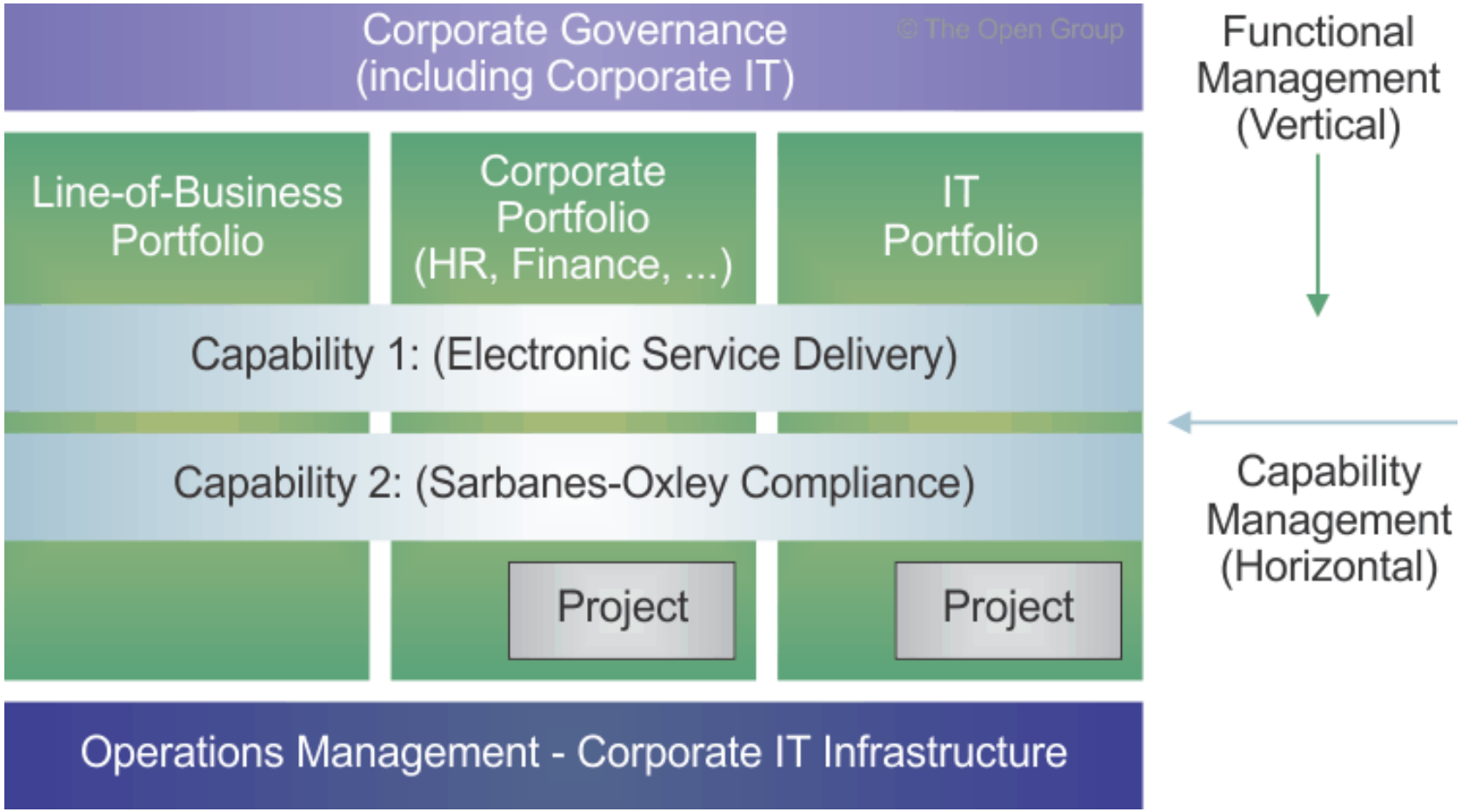
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Risk ID	Risk	Preliminary Risk			Mitigation	Residual Risk		
		Effect	Frequency	Impact		Effect	Frequency	Impact

Capability Based Planning

- Capability-based planning is a technique that focuses on the **planning**, **engineering** and **delivery** of strategic **business capabilities**
- It frames all phases of the architecture development in the context of **business outcomes**, clearly linking the IT vision, architectures (ABBs and SBBs), and the Implementation and Migration Plans with corporate strategic, business and line of business plans

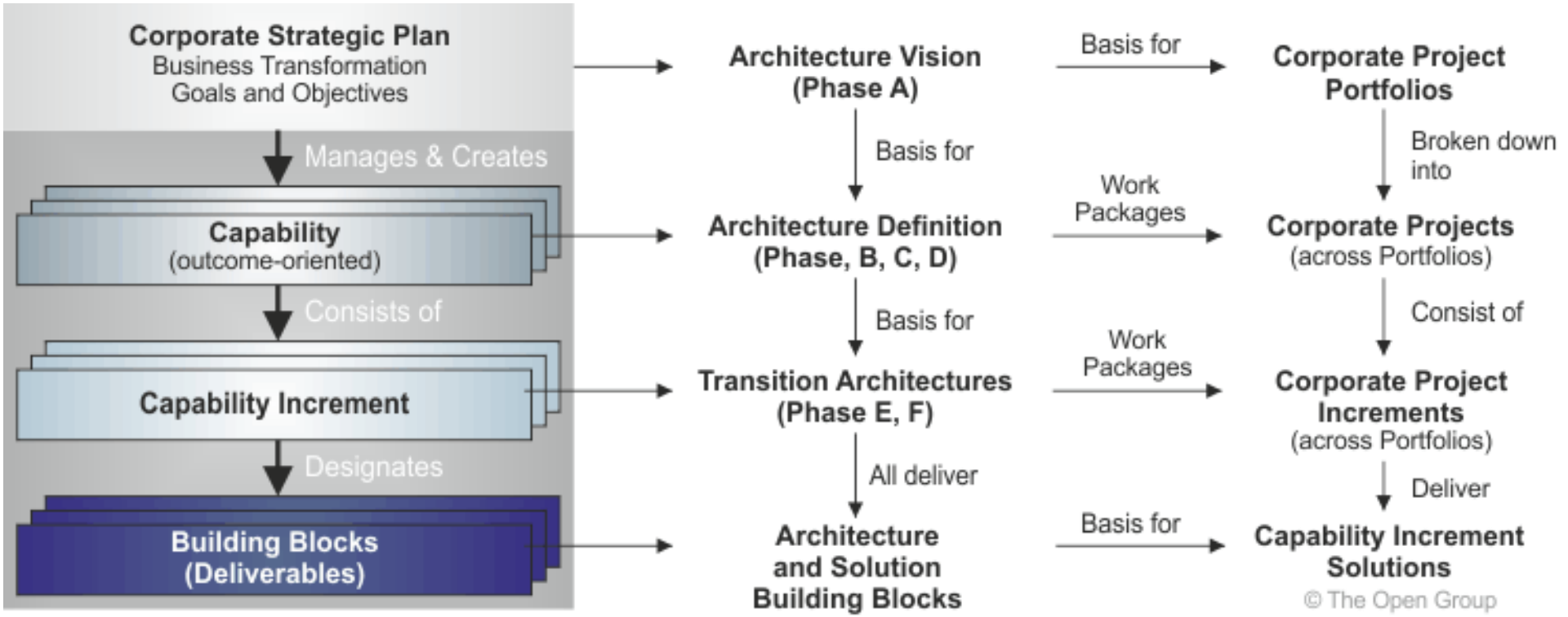
Capabilities



Capability Based Planning

- Capabilities are directly derived from the **corporate strategic plan**. They must satisfy the enterprise goals, objectives and strategies. Most organizations will also have an annual business plan
- All of the architectures will be expressed in terms of business outcomes and value
- **Phase A**: the corporate strategic direction must drive this
- **Phases B, C and D**: specific capabilities must be targeted for completion
- **Phase E**: the capability increments must drive this

Capability Based Planning



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Summary

This module has explained how to apply different techniques to help with the implementation of the architectures defined in the ADM phases, including

- Interoperability requirements
- Business transformation readiness factors
- How to perform an initial risk assessment
- How capability based planning can be applied